

Mathematical Physics

Solved Past Paper

Heat Equation · Wave Equation · D'Alembert · Separation of Variables

COMPOSED BY

Nadir Ali Khan

BS-IV Mathematics

TEACHER

Sir Darshan

Mathematical Physics

Exam: Final Examinations 2025 **Class:** BS-III
Total Marks: 50 (Attempt any **Three** questions) **Time:** 2 Hours
Date: Tuesday, June 02, 2025

Q.No. 1 — Heat Equation on a Circular Wire (Periodic BCs)

Find the temperature $u(x, t)$ in a thin wire on $[-2, 2]$ bent into a circle (ends $x = -2$ and $x = 2$ joined), satisfying:

$$k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad -2 < x < 2, \quad t > 0$$

$$u(-2, t) = u(2, t), \quad \frac{\partial u}{\partial x} \Big|_{x=-2} = \frac{\partial u}{\partial x} \Big|_{x=2}, \quad t > 0; \quad u(x, 0) = 1, \quad -2 < x < 2.$$

Solution**Step 1 — Assume a Fourier series solution.**

For periodic BCs on $[-L, L]$ with $L = 2$, the solution takes the form:

$$u(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{2}\right) + b_n \sin\left(\frac{n\pi x}{2}\right) \right] e^{-k(n\pi/2)^2 t}$$

Step 2 — Find Fourier coefficients from $u(x, 0) = 1$.

$$a_0 = \frac{1}{2} \int_{-2}^2 1 \, dx = \frac{1}{2}(4) = 2$$

$$a_n = \frac{1}{2} \int_{-2}^2 \cos\left(\frac{n\pi x}{2}\right) dx = \frac{1}{2} \left[\frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) \right]_{-2}^2 = \frac{1}{n\pi} [\sin(n\pi) - \sin(-n\pi)] = 0$$

$$b_n = \frac{1}{2} \int_{-2}^2 \sin\left(\frac{n\pi x}{2}\right) dx = 0 \quad (\text{odd integrand on symmetric interval})$$

Step 3 — Write the solution.

Since all $a_n = b_n = 0$ for $n \geq 1$ and $a_0 = 2$:

$$\boxed{u(x, t) = 1}$$

Physical interpretation: The initial temperature is uniform ($u = 1$ everywhere). With periodic (insulating) boundary conditions and no heat loss, the temperature remains uniformly 1 for all time $t > 0$.

Q.No. 2 — Vibrating String (Wave Equation)

The vibration of a string of length 1 (both ends fixed) is governed by:

$$4 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0$$

with $u(0, t) = u(1, t) = 0$, initial velocity $u_t(x, 0) = 0$, and initial deflection $u(x, 0) = x$. Find $u(x, t)$.

Solution

Step 1 — Identify wave speed.

Compare with $c^2 u_{xx} = u_{tt}$: here $c^2 = 4$, so $c = 2$.

Step 2 — Separation of Variables. Let $u(x, t) = X(x)T(t)$:

$$\frac{X''}{X} = \frac{T''}{4T} = -\lambda$$

Spatial problem: $X'' + \lambda X = 0$, $X(0) = X(1) = 0$.

Eigenvalues: $\lambda_n = n^2 \pi^2$, Eigenfunctions: $X_n = \sin(n\pi x)$, $n = 1, 2, 3, \dots$

Temporal problem: $T'' + 4n^2 \pi^2 T = 0$

$$T_n(t) = A_n \cos(2n\pi t) + B_n \sin(2n\pi t)$$

Step 3 — Apply initial condition $u_t(x, 0) = 0$.

$$u_t = \sum [-2n\pi A_n \sin(2n\pi t) + 2n\pi B_n \cos(2n\pi t)] \sin(n\pi x)$$

$$\text{At } t = 0: \sum 2n\pi B_n \sin(n\pi x) = 0 \Rightarrow B_n = 0.$$

General solution:

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos(2n\pi t) \sin(n\pi x)$$

Step 4 — Apply initial deflection $u(x, 0) = x$.

$$x = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \Rightarrow a_n = 2 \int_0^1 x \sin(n\pi x) dx$$

Integration by parts:

$$\begin{aligned} \int_0^1 x \sin(n\pi x) dx &= \left[-\frac{x \cos(n\pi x)}{n\pi} \right]_0^1 + \frac{1}{n\pi} \int_0^1 \cos(n\pi x) dx = -\frac{\cos(n\pi)}{n\pi} + 0 = \frac{(-1)^{n+1}}{n\pi} \\ a_n &= \frac{2(-1)^{n+1}}{n\pi} \end{aligned}$$

Step 5 — Final solution:

$$u(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cos(2n\pi t) \sin(n\pi x)$$

Q.No. 3 — D'Alembert's Formula

Define and use D'Alembert's formula to solve:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad -\infty < x < +\infty, \quad t > 0; \quad u(x, 0) = \sin x, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 1.$$

Solution**D'Alembert's Formula (Definition):**

For the wave equation $u_{tt} = c^2 u_{xx}$ on $(-\infty, +\infty)$ with initial conditions $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$, the unique solution is:

$$u(x, t) = \frac{f(x+ct) + f(x-ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

Identify parameters: $c = 1$, $f(x) = \sin x$, $g(x) = 1$.

Step 1 — Apply the formula:

$$u(x, t) = \frac{\sin(x+t) + \sin(x-t)}{2} + \frac{1}{2} \int_{x-t}^{x+t} 1 ds$$

Step 2 — Evaluate the integral:

$$\frac{1}{2} \int_{x-t}^{x+t} ds = \frac{1}{2} [(x+t) - (x-t)] = \frac{1}{2} (2t) = t$$

Step 3 — Simplify using sum-to-product identity:

$$\sin(x+t) + \sin(x-t) = 2 \sin x \cos t \quad \left[\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} \right]$$

$$\frac{\sin(x+t) + \sin(x-t)}{2} = \sin x \cos t$$

Step 4 — Final solution:

$$u(x, t) = \sin x \cos t + t$$

Verification:

- $u(x, 0) = \sin x \cos 0 + 0 = \sin x \checkmark$
- $u_t = -\sin x \sin t + 1$; $u_t(x, 0) = 0 + 1 = 1 \checkmark$
- $u_{xx} = -\sin x \cos t$; $u_{tt} = -\sin x \cos t$; $u_{xx} = u_{tt} \checkmark$

Q.No. 4 — Product Solution + Heat Equation in a Bar

(a) Find the product solution of $x \frac{\partial u}{\partial x} = y \frac{\partial u}{\partial y}$.

Solution

Assume $u(x, y) = X(x)Y(y)$. Substituting:

$$x X'(x) Y(y) = y X(x) Y'(y)$$

Separate variables (divide both sides by $X(x)Y(y)$):

$$\frac{x X'}{X} = \frac{y Y'}{Y} = k \quad (\text{separation constant})$$

Solve for $X(x)$: $\frac{X'}{X} = \frac{k}{x}$

$$\ln X = k \ln x + C_1 \Rightarrow X(x) = A x^k$$

Solve for $Y(y)$: $\frac{Y'}{Y} = \frac{k}{y}$

$$\ln Y = k \ln y + C_2 \Rightarrow Y(y) = B y^k$$

Product solution:

$$u(x, y) = C (xy)^k$$

where C and k are arbitrary constants. One can verify: $x \cdot C k (xy)^{k-1} y = y \cdot C k (xy)^{k-1} x \checkmark$.

(b) Find the temperature in a bar of length π , ends kept at zero, lateral surface insulated, with initial temperature $u(x, 0) = x(\pi - x)$.

Solution

The BVP:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < \pi, \quad t > 0; \quad u(0, t) = u(\pi, t) = 0; \quad u(x, 0) = x(\pi - x).$$

Step 1 — General solution (Fourier sine series):

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin(nx) e^{-n^2 t}$$

Step 2 — Find coefficients $b_n = \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin(nx) dx$.

Expand the integral:

$$I = \int_0^\pi (\pi x - x^2) \sin(nx) dx = \pi \int_0^\pi x \sin(nx) dx - \int_0^\pi x^2 \sin(nx) dx$$

Integral 1 (by parts, $u = x$, $dv = \sin(nx) dx$):

$$\int_0^\pi x \sin(nx) dx = \left[-\frac{x \cos(nx)}{n} \right]_0^\pi + \frac{1}{n} \int_0^\pi \cos(nx) dx = -\frac{\pi(-1)^n}{n} + 0 = \frac{\pi(-1)^{n+1}}{n}$$

Integral 2 (by parts, $u = x^2$, $dv = \sin(nx) dx$):

$$\int_0^\pi x^2 \sin(nx) dx = \left[-\frac{x^2 \cos(nx)}{n} \right]_0^\pi + \frac{2}{n} \int_0^\pi x \cos(nx) dx = -\frac{\pi^2(-1)^n}{n} + \frac{2}{n} \cdot \frac{(-1)^n - 1}{n^2}$$

where $\int_0^\pi x \cos(nx) dx = \frac{(-1)^n - 1}{n^2}$ (by parts).

Combining:

$$\begin{aligned} I &= \pi \cdot \frac{\pi(-1)^{n+1}}{n} - \left[-\frac{\pi^2(-1)^n}{n} + \frac{2((-1)^n - 1)}{n^3} \right] \\ &= \frac{\pi^2(-1)^{n+1}}{n} + \frac{\pi^2(-1)^n}{n} - \frac{2((-1)^n - 1)}{n^3} \\ &= \frac{\pi^2(-1)^n(-1 + 1)}{n} + \frac{2(1 - (-1)^n)}{n^3} = \frac{2(1 - (-1)^n)}{n^3} \end{aligned}$$

Step 3 — Evaluate b_n :

$$b_n = \frac{2}{\pi} \cdot \frac{2(1 - (-1)^n)}{n^3} = \frac{4(1 - (-1)^n)}{\pi n^3}$$

$$b_n = \begin{cases} 0 & n \text{ even} \\ \frac{8}{\pi n^3} & n \text{ odd} \end{cases}$$

Step 4 — Final solution:

$$u(x, t) = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{\sin((2n-1)x)}{(2n-1)^3} e^{-(2n-1)^2 t}$$

Note: At $t = 0$: $(8/\pi) \sum_{n=1}^{\infty} \sin((2n-1)x)/(2n-1)^3 = x(\pi - x)$ for $0 < x < \pi$ ✓ (standard Fourier sine result).